

Omraj Burnwal

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Education

M.Tech (Research)
B.Tech

IIT Bombay 2020–2023
IIT Ropar 2015–2019

Certifications

- Acquired **CompTIA Security+** certification, covering network security, cryptography, risk management, and incident response.
- Completed multiple hands-on learning paths on **TryHackMe** covering SOC, defensive security, and exploitation fundamentals.

Projects

SIEM Tools

- **Splunk:** Configured agents (Universal Forwarders) for robust log ingestion, utilized SPL (Search Processing Language) for advanced queries, and created dashboards and reports for real time security monitoring and data visualization.
- **Wazuh:** Deployed and configured the open source Wazuh Agent and Manager, leveraged the platform for host-based intrusion detection, and developed custom rules for file integrity monitoring and system security configuration assessment.

Live SOC Monitoring (TryHackMe)

January'26 – Present

Monitored real time security alerts, performed triage and investigated incidents in a simulated SOC environment.

- Phishing alert

Exploitation Basics

- Completed Exploitation Basics module with hands-on practice in identifying and exploiting common system and web vulnerabilities.
- Gained practical understanding of attack surfaces, reconnaissance, and initial access techniques used in real-world penetration testing.
- Worked with basic exploitation concepts including vulnerable services, misconfigurations, and insecure applications.
- Practiced using industry-standard tools such as Nmap, Metasploit, and Linux command-line utilities for vulnerability discovery and exploitation.

Defensive Security Tooling

Experiences

Research Assistant at Wadhvani Electronics Lab, IIT Bombay

- Assisted in designing and conducting digital and analog communication laboratory experiments.
- Guided and evaluated undergraduate students in both online and offline lab sessions.
- Managed administrative responsibilities for Communication and Electronics Design Labs.

NPTEL Workshop, IIT Bombay

- Helped college students to learn about microprocessor (8085) programming.
- Conducted labs for basic hands on practice.

Publication

Full-Duplex Relay Assisted Coded Cooperation, IEEE INDICON Conference

Paper

Previous Experience

Straggler Mitigation in Distributed Matrix Multiplication

June'24 - Nov'24

- Studied coded computation techniques for mitigating slow nodes in distributed systems.
- Analyzed classical non-learning based coded computation schemes.

Internship

- Worked on creating a simulated Indoor tracking system using RFID tags at **iD Tech**

Outreach and Other Responsibilities

Mentorship – IIT Ropar

- Mentored 5 undergraduate students at IIT Ropar.
- Conducted regular one-on-one and small-group meetings to help students acclimate to college life.

Fine Arts Club Head Coordinator

- Organized various painting and decoration contests at IIT Ropar
- Lead a team of around 20 people for decorating and designing the theme for cultural fest Zeitgeist-2016 and Zeitgeist-2017.

Academic Tutor - Edufeat

- Provided problem-solving support in Control Systems, Signals & Systems, Calculus, and Probability to college students.

Technical Skills

Programming: Python, MATLAB, C++

Tools: Splunk, Wazuh, Metasploit, Wireshark, Nmap, Burp Suite, Cisco Packet Tracer

OS: Kali linux, Ubuntu, and Windows

Interests

Video games, Anime, Movies, Music